

DeWM: Role-Specialized Latent Dynamics for Reconstruction-Free World Models

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Abstract

JEPA-based latent world models enable efficient planning by predicting compact future representations without pixel-level reconstruction. However, existing approaches typically model heterogeneous predictive factors through a homogeneous next-state prediction mechanism, which may bias compact representations toward globally predictive features while underrepresenting fine-grained spatial information and event-sensitive cues. We propose DeWM (**D**ecoupled **L**atent **W**orld **M**odel), a reconstruction-free world model that functionally decomposes the predictive latent state into a task-state component and a spatial-detail component. The former captures smooth macroscopic physical evolution and global task progression, whereas the latter preserves fine-grained geometric relations and event-sensitive changes associated with contact and abrupt displacement. To account for their distinct temporal characteristics, we introduce a unified role-specialized dynamics framework that jointly constrains action-conditioned latent velocities and the outcomes obtained by integrating the learned velocity fields. Within this framework, process-oriented dynamics emphasize the evolution of the task-state component, while outcome-oriented dynamics prioritize the accuracy of event-induced transitions in the spatial-detail component. This functional decomposition further enables configurable *Event-Evolution Planning Modes*, allowing planning preferences to be adjusted at inference time without retraining. Experiments across diverse navigation, manipulation, and continuous-control benchmarks demonstrate that DeWM improves spatially sensitive planning while preserving general dynamics modeling capability.

Introduction

Recent JEPA-based world models (Zhou et al. 2024; Garrido et al. 2024; Assran et al. 2023a, 2025; Sobal et al. 2025; Nam et al. 2026; Maes et al. 2026) have demonstrated that effective latent dynamics can be learned by directly predicting future representations, without requiring pixel-level reconstruction. By mapping observations into compact predictive embeddings, these models reduce the computational burden of future prediction while retaining task-related information.

Despite their efficiency, existing JEPA-based world models typically compress heterogeneous predictive factors into

a compact latent representation and model their evolution through a homogeneous next-state prediction mechanism. Such homogeneous treatment may bias the learned representation toward globally predictive features while underrepresenting fine-grained spatial information and event-sensitive cues (*i.e.*, contact and displacement). This limitation is problematic for robotic manipulation, where successful planning requires capturing both global task progression and localized spatial information. This motivates a natural question: *how can heterogeneous predictive factors be functionally organized and modeled with dynamics suited to their distinct roles in reconstruction-free JEPA-based world models?*

To this end, we propose DeWM, a reconstruction-free framework that functionally decomposes the predictive latent state into a task-state component and a spatial-detail component, and models them with role-specialized dynamics aligned with their respective temporal characteristics. Specifically, the task-state component captures global task progression, macroscopic physical evolution, and the overall consequences of actions, whose temporal transition is generally smooth. In contrast, the spatial-detail component preserves fine-grained geometric relations and event-sensitive changes, including contact and abrupt displacement. Because such changes are often temporally localized and non-smooth, accurately predicting the resulting latent state is more important than modeling their intermediate latent evolution.

In practice, DeWM comprises a shared ViT-based visual encoder and a next-latent-state predictor parameterized by a latent velocity field. The encoder maps each observation to a [CLS] token and a sequence of patch tokens. The [CLS] token is projected into a low-dimensional task-state component, whereas the patch tokens are spatially pooled and projected into a low-dimensional spatial-detail component. The predictor models the action-conditioned transitions of both components through a unified role-specialized dynamics framework. It directly estimates their component-wise latent velocities, whose numerical integration produces the predicted next latent state, thereby enabling constraints on both the velocity field and its integrated outcome. Within this framework, the task-state component is governed by *process-oriented dynamics*: velocity matching emphasizes the smooth macroscopic task evolution, while outcome attraction anchors the integrated transition to the target latent state. In contrast, the spatial-detail component is governed

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by *outcome-oriented dynamics*: weaker velocity matching maintains temporal coherence, whereas stronger outcome attraction emphasizes the accuracy of event-induced transitions. The two components are therefore differentiated by how process fidelity and outcome fidelity are balanced according to their functional roles. Finally, separate SigReg. regularization is further applied to these two components to maintain their distinct statistical structures. Although the components remain independent throughout prediction, their respective learning objectives jointly update the shared ViT encoder through backpropagation, enabling coordinated representation learning.

During deployment, the functional decomposition also exposes a configurable inference-time property, which we term *Event–Evolution Planning Modes*. A scalar coefficient modulates the contribution of the spatial-detail component, allowing the planner to operate under different preferences between macroscopic evolution consistency and event-sensitive spatial reasoning. This provides a practical means of adapting planning preferences without retraining: contact-rich or spatially demanding tasks can adopt an event-enhanced mode, whereas tasks requiring long-horizon motion consistency can adopt an evolution-focused mode. Such component-level control is difficult to achieve in existing JEPA-based world models, whose homogeneous representations do not expose functionally distinct factors for independent adjustment.

Our contributions are summarized as follows:

- We propose DeWM, a reconstruction-free latent world model that functionally decomposes heterogeneous predictive factors into compact, role-specialized components, addressing the lack of functional organization in existing JEPA-based world models.
- We introduce role-specialized dynamics modeling to account for the heterogeneous temporal properties of latent factors. Within a unified dynamics framework, *process-oriented* dynamics characterize the smooth evolution of the task-state component, whereas *outcome-oriented* dynamics emphasize event-sensitive transitions in the spatial-detail component.
- Extensive experiments across navigation, manipulation, and continuous-control benchmarks demonstrate that our DeWM improves event-sensitive planning while preserving general dynamics modeling capacity.

Related Work

Latent World Models. World models (Finn and Levine 2017; Ha and Schmidhuber 2018; Kaiser et al. 2019; Schrittwieser et al. 2020; Bruce et al. 2024; Hafner et al. 2019a,b, 2020, 2025) learn predictive environment dynamics to support reasoning, planning, and control. Early approaches commonly relied on generative objectives that reconstruct future observations or model their likelihoods (Micheli, Alonso, and Fleuret 2022; Alonso et al. 2024). Although effective, such objectives are computationally demanding and may allocate substantial modeling capacity to visual details irrelevant to control. JEPA-based models instead learn predictive representations by predicting targets in latent space (Assran et al. 2023b; Bardes et al. 2024). This reconstruction-free

paradigm has recently been extended to action-conditioned world modeling for visual planning (Sobal et al. 2025; Zhou et al. 2024; Maes et al. 2026). Recent studies further investigate how predictive factors can be organized within the latent space. Sub-JEPA (Zhao et al. 2026) applies Gaussian regularization over multiple random subspaces, relaxing the full-space isotropic prior while retaining its anti-collapse effect. SD-JEPA (Thil et al. 2026) decomposes the representation into orthogonal progression and content subspaces and assigns them distinct objectives. Nevertheless, both approaches retain a full-latent next-state prediction objective and induce specialization primarily through regularization or auxiliary objectives, without explicitly modeling the distinct transition characteristics of different predictive factors. Our work instead organizes global task-state and local spatial-detail factors according to their planning roles and models them with *process-oriented* and *outcome-oriented* dynamics.

Spatial Representations for Robotic Dynamics. Successful robotic manipulation requires representations that preserve fine-grained spatial information (*i.e.*, geometric relations and contact-relevant regions). Compact embeddings are efficient but may suppress spatial details, whereas dense patch-level or object-centric representations retain richer spatial structure at increased computational or architectural cost (Zhou et al. 2024; Wang et al. 2026). However, retaining spatial features alone neither specifies their predictive role nor determines how their dynamics should be modeled. DeWM compresses global and patch-level ViT features into low-dimensional task-state and spatial-detail components, respectively. Our focus is therefore not merely on retaining additional spatial information, but on assigning global and local predictive factors distinct functional roles and modeling their dynamics within a reconstruction-free world model.

Flow Matching for Latent Dynamics. Flow matching learns continuous velocity fields along prescribed transition paths (Lipman et al. 2022; Liu, Gong, and Liu 2022; Tong et al. 2023) and has been applied to image, video, and action generation (Dao et al. 2023; Li et al. 2026b; Murillo-González and Liu 2025). Recently, flow-based generative modeling has been incorporated into world models to predict future states in latent space (Li et al. 2026a; Ye et al. 2026; Tian et al. 2026). We organize the predictive latent state into functionally distinct task-state and spatial-detail components, and associate each component with a different balance of process fidelity and outcome fidelity. Within this role-specialized formulation, *process-oriented* dynamics emphasize the smooth evolution of macroscopic physics and task progression, whereas *outcome-oriented* dynamics prioritize event-induced changes in spatial details.

Method

Within a reconstruction-free world-modeling framework, our goal is to organize the predictive latent state into functionally distinct, planning-relevant components and explicitly model their respective dynamics. This enables the world model to capture both coherent macroscopic task evolution and accurate event-induced outcomes for goal-conditioned planning.

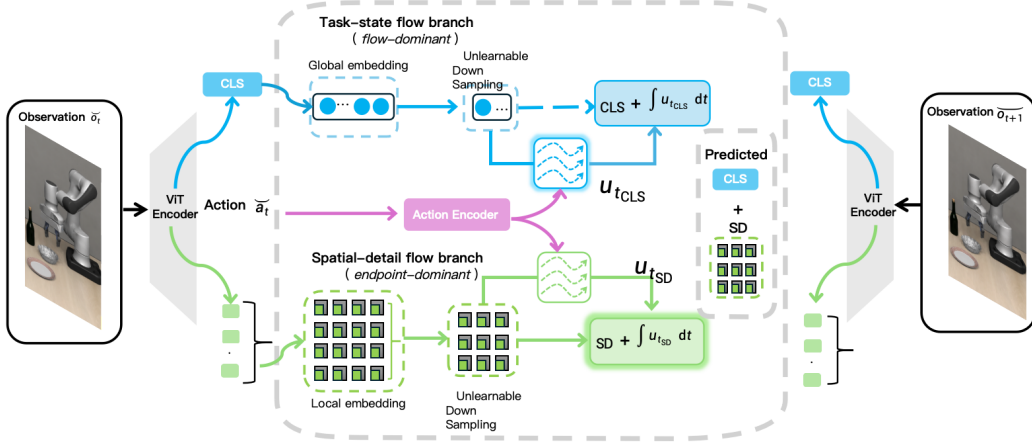


Figure 1: Overview of our DeWM.

Framework Overview

As illustrated in Fig. 1, our proposed DeWM follows an action-conditioned JEPA framework and consists of a shared ViT-based visual encoder $\mathcal{E}_\phi$, an action encoder $\mathcal{E}_a$, and a next-latent-state predictor Φ_θ induced by a latent velocity-field network $\mathcal{V}_\theta$. Given a transition tuple (o_t, a_t, o_{t+1}) , the visual encoder produces the current and future latent states z_t and z_{t+1} , respectively. Conditioned on the current latent state and the encoded action, the velocity network predicts component-wise latent velocity fields that describe the continuous evolution from z_t toward z_{t+1} . Numerically integrating these velocity fields yields the one-step transition $\hat{z}_{t+1} = \Phi_\theta(z_t, a_t)$. The visual encoder, action encoder, and dynamics model are jointly optimized entirely in latent space, without pixel-level reconstruction.

Functionally Decoupled Latent State

A shared ViT encoder $\mathcal{E}_\phi$ maps an observation o_t to a [CLS] token $h_t^{[\text{CLS}]}$ and a sequence of patch tokens H_t^{patch} :

$$[h_t^{[\text{CLS}]}, H_t^{\text{patch}}] = \mathcal{E}_\phi(o_t). \quad (1)$$

Task-State Component. The [CLS] token is mapped to a compact task-state embedding z_t^{ts} :

$$z_t^{\text{ts}} = g(h_t^{[\text{CLS}]}) , \quad (2)$$

where g is a lightweight projection head. The task-state component summarizes global task progression and macroscopic physical evolution, providing a compact predictive representation for determining whether a candidate action induces coherent evolution toward the goal.

Spatial-Detail Component. To retain localized geometric information, the patch-token sequence is restored to its two-dimensional arrangement and pooled into a fixed $k \times k$ spatial grid:

$$r_t = g_s(\text{Pool}_{k \times k}(\text{Reshape}(H_t^{\text{patch}}))) , \quad (3)$$

where g_s denotes a point-wise spatial projection head. The spatial-detail embedding z_t^{sd} is obtained by flattening the resulting grid r_t :

$$z_t^{\text{sd}} = \text{Flatten}(r_t). \quad (4)$$

Unlike the task-state component, the spatial-detail component preserves the indexed spatial organization of the pooled patch features. It is therefore designed to retain local geometric relations and event-sensitive changes associated with contact and scene structure.

The complete predictive latent state is

$$z_t = [z_t^{\text{ts}}; z_t^{\text{sd}}] . \quad (5)$$

The two components are produced by separate projection heads while sharing the same visual encoder $\mathcal{E}_\phi$. Their learning signals therefore jointly update the shared visual backbone while preserving distinct functional roles.

Role-Specialized Latent Dynamics

The two latent components exhibit distinct temporal characteristics under action-conditioned dynamics. We model the task-state component with *process-oriented dynamics*, which emphasize the evolution path of its latent transition, whereas the spatial-detail component is modeled with *outcome-oriented dynamics*, which prioritize its final latent state resulting from the transition. Both are implemented within a unified velocity-field formulation, but differ in the dynamics property emphasized for their own functional roles.

Process-Oriented Dynamics. Given two consecutive latent states z_t and z_{t+1} , we build a prescribed latent bridge:

$$z_t^{(\tau)} = (1 - \tau)z_t + \tau z_{t+1}, \quad \tau \sim \mathcal{U}(0, 1). \quad (6)$$

The corresponding transition velocity is

$$[u_t^{\text{ts}}; u_t^{\text{sd}}] = u_t = \frac{dz_t^{(\tau)}}{d\tau} = z_{t+1} - z_t. \quad (7)$$

We parameterize its latent dynamics using an velocity-field network $\mathcal{V}_\theta$:

$$[v_\tau^{\text{ts}}; v_\tau^{\text{sd}}] = v_\tau = \mathcal{V}_\theta \left(z_t^{(\tau)}, a_t, \tau \right). \quad (8)$$

Here, $\mathcal{V}_\theta$ constitutes a unified dynamics model that predicts the velocities of the two latent components. The process objectives identify the action-conditioned velocity field at sampled intermediate latent states:

$$\begin{aligned} \mathcal{L}_{\text{pro}}^{\text{ts}} &= \mathbb{E}_{t,\tau} \left[\left\| v_\tau^{\text{ts}} - u_t^{\text{ts}} \right\|_2^2 \right], \\ \mathcal{L}_{\text{pro}}^{\text{sd}} &= \mathbb{E}_{t,\tau} \left[\left\| v_\tau^{\text{sd}} - u_t^{\text{sd}} \right\|_2^2 \right]. \end{aligned} \quad (9)$$

The prescribed bridge provides an auxiliary path for identifying the local evolution field and should not be assumed to recover the unobserved physical micro-trajectory between two consecutive observations.

Outcome-Oriented Dynamics. While the process objectives characterize the local transition direction, goal-conditioned planning requires accurate prediction of the latent state reached after applying an action. We therefore additionally model the transition outcome by integrating the predicted velocity field. Starting from the current latent state z_t , we approximate the one-step transition using K Euler integration steps. At first, we initialize the integration state $\hat{z}_t^{(0)}$ and define the step size $\Delta\tau$ and integration time τ_k as

$$\hat{z}_t^{(0)} = z_t, \Delta\tau = \frac{1}{K}, \tau_k = \frac{k}{K}. \quad (10)$$

At each integration step, the current predicted latent state is fed back into $\mathcal{V}_\theta$ to estimate the local action-conditioned velocity:

$$\hat{z}_t^{(k+1)} = \hat{z}_t^{(k)} + \Delta\tau \mathcal{V}_\theta \left(\hat{z}_t^{(k)}, a_t, \tau_k \right). \quad (11)$$

After K steps, the resulting latent state defines the predicted outcome of the one-step action-conditioned transition:

$$\begin{aligned} [\hat{z}_{t+1}^{\text{ts}}; \hat{z}_{t+1}^{\text{sd}}] &= \hat{z}_{t+1} = \Phi_\theta(z_t, a_t) \\ &= z_t + \Delta\tau \sum_{k=0}^{K-1} \mathcal{V}_\theta \left(\hat{z}_t^{(k)}, a_t, \tau_k \right). \end{aligned} \quad (12)$$

We model the outcome of each one-step transition by matching the integrated prediction to the next latent state z_{t+1} :

$$\begin{aligned} \mathcal{L}_{\text{out}}^{\text{ts}} &= \mathbb{E}_t \left[\left\| \hat{z}_{t+1}^{\text{ts}} - z_{t+1}^{\text{ts}} \right\|_2^2 \right], \\ \mathcal{L}_{\text{out}}^{\text{sd}} &= \mathbb{E}_t \left[\left\| \hat{z}_{t+1}^{\text{sd}} - z_{t+1}^{\text{sd}} \right\|_2^2 \right]. \end{aligned} \quad (13)$$

Unlike direct next-state regression, the predicted outcome is generated by repeatedly querying and integrating the learned velocity field used for *process-oriented* dynamics. The outcome objectives therefore constrain the complete transition operator Φ_θ rather than introducing an additional endpoint predictor. Moreover, because each intermediate input $\hat{z}_t^{(k)}$ is generated by the model itself, the integrated transition directly evaluates the velocity field along its predicted rollout.

Joint Modeling of Heterogeneous Dynamics. The process and outcome objectives characterize complementary aspects of an action-conditioned latent transition, but their relative importance differs across the two components. We jointly learn these role-specialized dynamics using

$$\mathcal{L}_{\text{dyn}} = \mathcal{L}_{\text{pro}}^{\text{ts}} + \alpha \mathcal{L}_{\text{out}}^{\text{ts}} + \beta \mathcal{L}_{\text{pro}}^{\text{sd}} + \mathcal{L}_{\text{out}}^{\text{sd}}. \quad (14)$$

The task-state component places greater emphasis on identifying the action-conditioned evolution process, whereas the spatial-detail component places greater emphasis on identifying the resulting latent state integration:

$$\alpha, \beta < 1. \quad (15)$$

The complete training objective is

$$\mathcal{L} = \mathcal{L}_{\text{dyn}} + \lambda(\mathcal{L}_{\text{reg}}^{\text{ts}} + \mathcal{L}_{\text{reg}}^{\text{sd}}). \quad (16)$$

Here, separate SigReg. avoids forcing the two components into a single homogeneous statistical structure.

Goal-Conditioned Planning

Given the current latent state z_t and a candidate action sequence with planning horizon H ,

$$\mathbf{a}_{t:t+H-1} = (a_t, a_{t+1}, \dots, a_{t+H-1}), \quad (17)$$

the latent rollout is generated by our action-conditioned predictor Φ_θ recursively, yielding

$$\hat{z}_{t+1}, \hat{z}_{t+2}, \dots, \hat{z}_{t+H}. \quad (18)$$

For a goal observation o_g , the shared visual encoder $\mathcal{E}_\phi$ and the two projection heads produce the corresponding goal task-state and spatial-detail embeddings:

$$z_g = [z_g^{\text{ts}}; z_g^{\text{sd}}]. \quad (19)$$

The candidate action sequence is then evaluated by comparing the predicted terminal latent state $\hat{z}_{t+H}$ with the goal:

$$\mathcal{J}(\mathbf{a}_{t:t+H-1}) = \mathcal{D}(\hat{z}_{t+H}^{\text{ts}}, z_g^{\text{ts}}) + \eta \mathcal{D}(\hat{z}_{t+H}^{\text{sd}}, z_g^{\text{sd}}), \quad (20)$$

where $\mathcal{D}(\cdot, \cdot)$ denotes the latent distance functions. Following prior latent-world-model planning protocols (Sobal et al. 2025; Maes et al. 2026), MPChuqibin selects the candidate action sequence that minimizes Eq. (20) and replans after receiving the next observation.

Event-Evolution Planning Modes. During inference, the functional decomposition exposes a configurable planning property through the coefficient η in Eq. (20), which modulates the contribution of the spatial-detail component to the planning objective. A smaller η places greater emphasis on global task progression and macroscopic physical evolution, whereas a larger η increases the influence of event-induced changes. Different values of η therefore instantiate different *Event-Evolution Planning Modes* without retraining the encoder or dynamics predictor. We use $\eta = 0.25$ as the default balanced setting in the main experiments.

Table 1: **Main Results** across diverse benchmarks, comparing the planning performance of DeWM with existing baselines.

Method	Two-Room	Push-T	OGBench		DeepMind Control suite		
			Cube	Scene	Reacher	Finger	Humanoid
GCIQL (Kostrikov, Nair, and Levine 2021)	100.00	20.00	64.00	57.00	1.67	83.00	27.00
GCIVL (Park et al. 2025)	100.00	33.00	56.00	55.33	1.67	87.00	27.33
DINO-WM (Zhou et al. 2024)	100.00	74.00	86.00	54.33	79.00	87.00	20.00
PLDM (Sobal et al. 2025)	97.00	78.00	65.00	52.33	78.00	88.00	18.67
LeWM (Maes et al. 2026)	84.33 $\pm$ 4.23	84.67 $\pm$ 6.53	67.33 $\pm$ 5.01	49.67 $\pm$ 4.46	82.67 $\pm$ 4.42	88.67 $\pm$ 5.61	18.33 $\pm$ 3.67
Sub-JEPA (Zhao et al. 2026)	95.00 $\pm$ 2.76	89.00 $\pm$ 5.33	76.33 $\pm$ 5.99	50.67 $\pm$ 5.47	84.00 $\pm$ 4.00	87.67 $\pm$ 5.28	19.33 $\pm$ 4.67
DeWM (Ours)	99.00 $\pm$ 1.00	93.67 $\pm$ 2.94	86.00 $\pm$ 3.65	58.33 $\pm$ 6.38	85.00 $\pm$ 3.58	90.00 $\pm$ 4.90	20.33 $\pm$ 4.27

Experiments

Experimental Setup

Benchmarks. Following the protocol of LeWM (Maes et al. 2026), we consider four previously evaluated tasks, including Two-Room (Sobal et al. 2025), Push-T (Zhou et al. 2024), OGBench Cube (Park et al. 2025), and Reacher from the DeepMind Control suite (Tassa et al. 2018). To further assess the representation ability of world models under more complex scene structures and dynamics, we additionally introduce three more challenging tasks: OGBench Scene, together with Finger and Humanoid from the DeepMind Control suite. These benchmarks span navigation, motion planning, and manipulation across both 2D and 3D scenes, providing a comprehensive testbed for evaluating world models under goal-conditioned control.

Baselines. Similar to LeWM (Maes et al. 2026), we compare our DeWM against both goal-conditioned policy baselines and JEPA-based world-model baselines. The policy baselines comprise GCIQL (Kostrikov, Nair, and Levine 2021) and GCIVL (Park et al. 2025), which serve as direct references to assess whether explicit world modeling offers additional advantages over purely behavior-level learning in the evaluated tasks. On the world-model side, we select PLDM (Sobal et al. 2025), DINO-WM (Zhou et al. 2024), LeWM (Maes et al. 2026), and Sub-JEPA (Zhao et al. 2026), all of which are designed to learn latent dynamics directly from pixel observations with built-in mechanisms against representation collapse. Among them, DINO-WM leverages a pretrained DINOv2 encoder to enhance dynamics modeling with off-the-shelf visual representations, while Sub-JEPA imposes the SigReg objective on latent subspaces, offering a more granular regularization scheme for learning semantically structured representations.

Planning Performance

For tasks that have been evaluated in LeWM (Maes et al. 2026) and Sub-JEPA (Zhao et al. 2026) (including Two-Room, Push-T, Cube, and Reacher), we directly cite the reported success rates of existing baselines when available. For the three newly introduced tasks (*i.e.*, Scene, Finger, and Humanoid), all compared methods are evaluated under an identical protocol. To account for stochastic variability, we

evaluate LeWM, Sub-JEPA, and our DeWM across six random seeds and present the mean and standard deviation of planning success rates. All results are compiled in Tab. 1. Overall, our DeWM achieves the best or highly competitive planning performance across most evaluated tasks.

On event-centric tasks such as Two-Room and Push-T, DeWM substantially outperforms LeWM, achieving approximately 10 percentage points higher on both tasks. This suggests that preserving spatial-detail information is beneficial for modeling event-induced transitions, particularly when successful planning hinges on accurately capturing local interactions or critical state changes. On the visually demanding OGBench tasks, DeWM further demonstrates strong advantages over latent world-model baselines. In Cube, DeWM improves upon LeWM and Sub-JEPA by 18% and 9% respectively, achieving performance comparable to DINO-WM. In the more challenging Scene task, DeWM achieves the highest success rate, improving over LeWM from 49.67% to 58.33%. These results suggest that modeling spatial details enables the latent representation to preserve more accurate geometric and positional information, which is important for robotic manipulation. On DeepMind Control suite, DeWM also delivers competitive performance under continuous dynamics. While the improvements on these lower-level motor-control tasks are more modest, the results still suggest that DeWM maintains competitive performance on continuous-control tasks while improving spatially sensitive capacity.

Spatial-Detail Benefits

Planning success measures whether a learned world model supports goal-conditioned control, but it does not directly reveal what information is encoded in the latent representation. We therefore analyze whether the spatial-detail component captures localized and event-sensitive information beyond the task-state component.

Event-Sensitive Perturbation Analysis. Given an original trajectory, we introduce a controlled perturbation only at step t while keeping the remaining observations unchanged. We consider three perturbation types on Push-T: a block-angle kick, a block-position nudge, and an agent-position nudge. As shown in Fig. 3, the spatial-detail component exhibits consistently stronger per-dimension responses to event-sensitive geometric perturbations than both the task-state

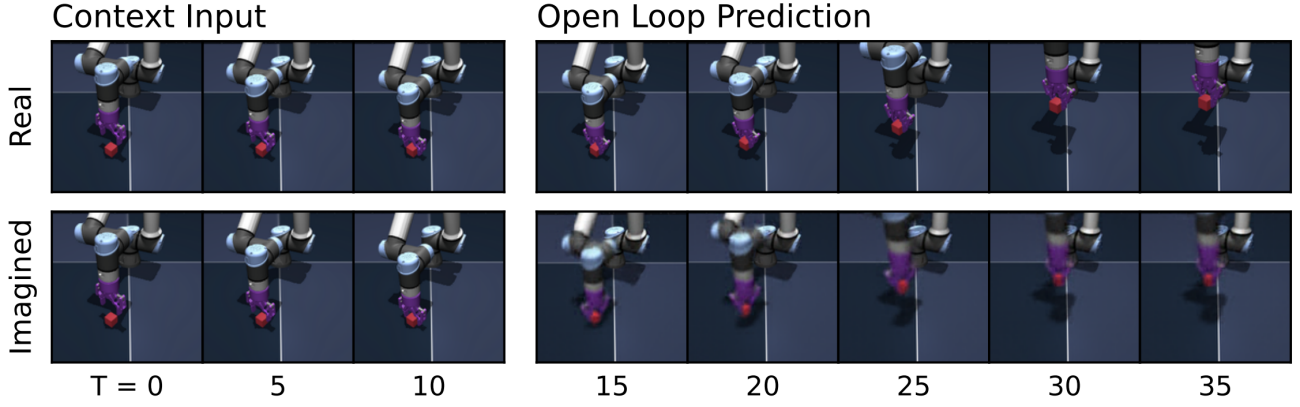


Figure 2: **Qualitative Comparison of Decoded Latent Rollouts.** The three rows show the ground-truth observations, decoded latent predictions from LeWM, and decoded latent predictions from DeWM, respectively. All predictions are generated under the same action sequence and visualized using a separately trained image decoder.

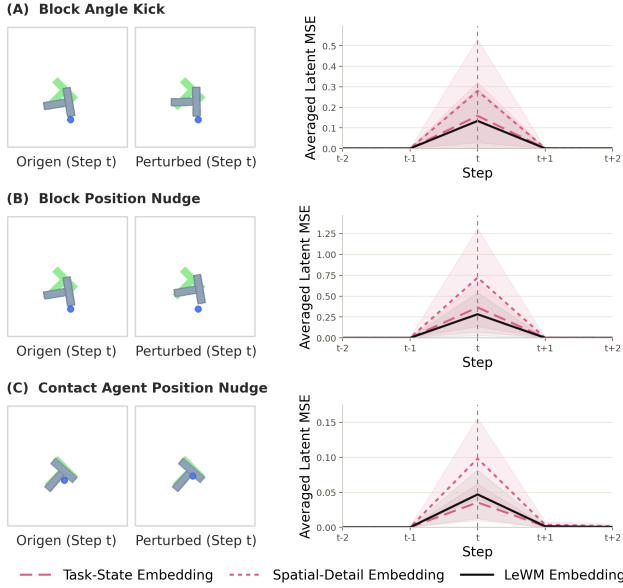


Figure 3: **Event-Sensitive Perturbation Analysis on Push-T.** The curves report the averaged MSE per latent dimension between the original and perturbed states, while the shaded regions denote variability across runs.

component and the homogeneous representation learned by LeWM. In contrast, the task-state component changes more conservatively, consistent with its role in capturing smooth macroscopic task evolution. Together, these response patterns support the intended functional specialization of the two latent components.

Physical State Probing. Following the probing protocol of LeWM (Maes et al. 2026), we freeze the visual encoder and train a lightweight MLP probe on Push-T to predict three physical variables: agent location, block location, and block angle. As shown in Table 2, DeWM consistently preserves

Table 2: **Physical State Probing on Push-T.** A lightweight MLP probe is trained on frozen latent representations to predict task-relevant physical variables.

Property	Method	MSE ↓	r ↑
Agent Loc.	PLDM (Sobal et al. 2025)	0.014	0.993
	DINO-WM (Zhou et al. 2024)	0.003	0.999
	LeWM (Maes et al. 2026)	0.004	0.998
	DeWM	0.003	0.999
Block Loc.	PLDM (Sobal et al. 2025)	0.011	0.994
	DINO-WM (Zhou et al. 2024)	0.002	0.999
	LeWM (Maes et al. 2026)	0.001	0.999
	DeWM	0.001	0.999
Block Angle	PLDM (Sobal et al. 2025)	0.056	0.972
	DINO-WM (Zhou et al. 2024)	0.009	0.995
	LeWM (Maes et al. 2026)	0.021	0.990
	DeWM	0.011	0.992

highly decodable physical information. It matches the best result for agent location and block location, while substantially reducing the block-angle prediction error of LeWM from 0.021 to 0.011. Although DINO-WM achieves the lowest error for block angle, DeWM remains competitive despite learning its visual representation end to end without relying on frozen pretrained features. These results suggest that introducing the spatial-detail embedding improves the retention of geometric and positional information without impairing the global task-state representation.

Qualitative Visualization. We further visualize the latent predictions of LeWM and DeWM using a separately trained image decoder. Specifically, each world model recursively predicts future latent states under the same action sequence, which are then decoded into image sequences for qualitative comparison. The decoder is used only for visualization and does not participate in world-model training or downstream planning. Fig. 2 illustrates that, compared with

Table 3: Success rates (%) under different inference-time Event–Evolution Planning Modes of DeWM.

Planning Mode	Two-Room	Push-T	Cube	Scene
Evolution-Focused ($\eta = 0$)	97.67	92.00	83.67	51.00
Balanced ($\eta = 0.25$)	99.00	93.67	86.00	58.33
Event-Enhanced ($\eta = 1$)	94.00	95.33	84.33	58.33

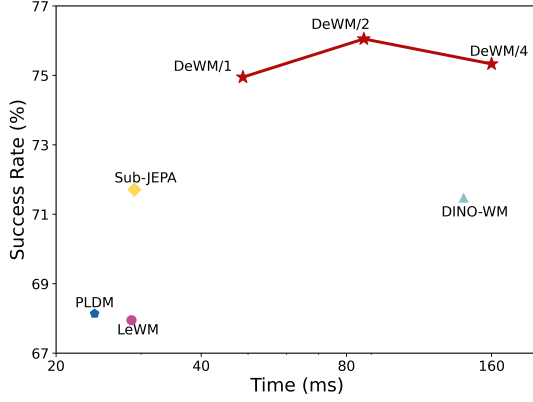


Figure 4: **Performance vs. Efficiency.** "Success Rate" is averaged over all tasks in Tab. 1, while "Time" denotes the average latency required to produce one action. DeWM/($K \in \{1, 2, 4\}$) uses K integration steps per latent transition.

LeWM, DeWM produces clearer and more temporally coherent decoded predictions, suggesting that the spatial-detail embedding preserves local geometric information that is easily attenuated in a single compact latent representation.

Event–Evolution Planning Modes

We further study how different inference-time modes affect planning performance. The Evolution-Focused mode assigns greater importance to the task-state embedding, whereas the Event-Enhanced mode emphasizes the spatial-detail embedding. The Balanced mode maintains an intermediate weighting between the two components. As shown in Tab. 3, the Balanced mode achieves the strongest overall performance, including the best results on Two-Room and Cube. In contrast, the Event-Enhanced mode performs best on Push-T and matches the best result on Scene, indicating that stronger spatial-detail guidance is particularly beneficial for interaction-intensive tasks involving contact and local geometric changes. These results demonstrate that DeWM can achieve better performance by adapting its planning mode to task characteristics at inference time, without retraining.

Ablation on Spatial-Detail Injection

Efficiency. As Fig. 4 shows, DeWM achieves a favorable performance–efficiency trade-off, substantially outperforming LeWM and Sub-JEPA while remaining considerably more efficient than DINO-WM. Notably, even the single-step variant of DeWM achieves strong planning performance, in-

Table 4: **Ablation of Supervision Strategy** on the Two-Room task. Each configuration is reported in the order of task-state supervision followed by spatial-detail supervision.

Representation	Task-State Dynamics	Spatial-Detail Dynamics	Two-Room	Push-T
Homogeneous Representation & Homogeneous Dynamics				
Homogeneous	Pure Process		-	87.67
Homogeneous	Pure Outcome		-	-
Decoupled Representation & Decoupled Dynamics				
Decoupled	Pure Outcome	Pure Outcome	59.33	-
Decoupled	Pure Outcome	Pure Process	67.33	27.67
Decoupled	Pure Process	Pure Outcome	92.67	-
Decoupled	Pure Process	Pure Process	95.00	-
Decoupled Representation & Role-Specialized Dynamics				
Decoupled	Process-Oriented	Outcome-Oriented	99.00	93.67

dicating that the gains of our method do not rely on computationally expensive numerical integration.

Supervision. We investigate whether the two latent components benefit from dynamics supervision aligned with their respective temporal characteristics. As shown in Tab. ??, applying outcome-only supervision to both components leads to the lowest success rate, indicating that direct next-state matching alone is insufficient for learning coherent action-conditioned transitions. Replacing the task-state supervision with the process objective produces a substantial improvement, from 67.33% for Outcome–Process to 92.67% for Process–Outcome. This result confirms that the task-state component benefits strongly from modeling the intermediate evolution of macroscopic task states. Using process supervision for both components further improves the success rate to 95.00%, suggesting that transition-direction modeling also provides useful temporal coherence for the spatial-detail component. Nevertheless, the best performance of 99.00% is achieved by the proposed role-specialized supervision, which combines process-dominant dynamics for the task-state component with outcome-dominant dynamics for the spatial-detail component. These results demonstrate that process and outcome supervision are complementary, and that balancing them according to the functional roles of the two components is more effective than applying a homogeneous supervision strategy.

Architecture Design. We investigate four key design choices in our DeWM, with results summarized in Tab. 5 and Fig. 4. **1) Regularization Strategy.** Separate SigReg. generally outperforms joint SigReg., suggesting that branch-specific constraints better preserve the functional distinction between the task-state and spatial-detail embeddings. **2) Down-Sampling Strategy.** Fixed average pooling consistently outperforms learnable self-attention under the same embedding dimensions, indicating that fixed spatial aggregation is sufficient for preserving local information. **3) Embedding Dimension.** The compact 64-64 configuration achieves the highest success rate with the lowest computational cost. Within the complexity range of the evaluated tasks, increasing the embedding dimensions brings no consistent performance gains while substantially increasing FLOPs. **4) In-**

Table 5: **Ablation of Architecture Design** on the Two-Room task. "TS-Dim" and "SD-Dim" denote the dimensions of the task-state and spatial-detail components, respectively. "Joint" applies SigReg to the concatenated embedding, whereas "Separate" regularizes them independently.

Down-Sampling Strategy	TS-Dim	SD-Dim	SigReg.	SR $\uparrow$ (%)	FLOPs $\downarrow$ (M)	Time $\downarrow$ (ms)
Fixed AvgPool	64	64	Joint	98.33	43.12	98.62
Fixed AvgPool	64	64	Separate	99.00	43.12	99.67
Fixed AvgPool	128	128	Joint	97.33	90.62	103.24
Fixed AvgPool	128	128	Separate	97.33	90.62	103.24
Fixed AvgPool	192	128	Joint	96.67	116.40	101.19
Fixed AvgPool	192	128	Separate	98.67	116.40	102.21
Learnable Self-Attn	192	128	Joint	95.33	116.40	102.11
Learnable Self-Attn	192	128	Separate	96.67	116.40	103.26

tegration Steps. Fig. 4 shows that increasing K from 1 to 2 improves success, whereas $K = 4$ adds latency without further gains. We therefore use $K = 2$ by default.

Conclusion

We presented DeWM, a reconstruction-free latent world model that functionally decomposes predictive representations into task-state and spatial-detail components. Rather than modeling heterogeneous predictive factors through a homogeneous next-state prediction mechanism, DeWM associates the two components with role-specialized dynamics. Process-oriented dynamics emphasize coherent macroscopic task evolution, whereas outcome-oriented dynamics prioritize the accuracy of localized, event-induced spatial transitions. This decomposition also exposes configurable Event-Evolution Planning Modes, enabling the relative contribution of spatial-detail information to be adjusted at inference time without retraining.

Experiments across navigation, robotic manipulation, and continuous-control benchmarks demonstrate that DeWM improves spatially sensitive planning while maintaining competitive general control performance. Perturbation analysis, physical-state probing, qualitative rollouts, and ablation studies further support the functional distinction between the two latent components and the effectiveness of their differentiated dynamics supervision. Future work may extend the current predefined decomposition toward automatically discovering task-dependent predictive roles and adaptively selecting planning modes from the current environment state.

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